

BALMACEDA, ChileWMO: **858740**Lat: **45.9128S**Lon: **71.6942W**Elev: **516**StdP: **95.28**Time Zone: **-4.00 (CHL)**Period: **96-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-11.3	-8.1	-13.0	1.3	-10.8	-9.9	1.7	-6.2	15.6	5.4	14.3	4.8	0.6	260	0.927

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.2	25.9	14.6	23.1	13.5	20.8	12.5	15.2	23.9	14.1	21.4	13.1	19.4	6.7	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
11.5	9.0	17.0	10.6	8.5	16.4	9.8	8.0	15.7	44.1	24.6	41.1	21.5	38.4	19.5	19.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		DB	WB
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	14.7	13.5	-15.3	28.2	3.9	3.0	-18.2	30.4	-20.5	32.1	-22.7	33.8	-25.6	36.0		
(5)			-15.6	16.7	4.0	1.2	-18.5	17.5	-20.8	18.2	-23.0	18.9	-25.9	19.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	12.6	12.3	10.2	7.3	4.1	1.4	1.0	2.8	5.1	7.3	8.9	11.3		
	DBStd	5.06	3.15	3.07	2.84	3.06	3.49	3.84	4.10	2.91	2.57	2.46	2.71	3.15		
	HDD10.0	1397	11	9	31	92	185	257	279	223	148	89	51	23		
	HDD18.3	4141	181	170	251	332	442	507	537	481	397	343	282	218		
	CDD10.0	301	90	74	38	9	2	0	0	0	1	5	18	63		
	CDD18.3	3	1	1	0	0	0	0	0	0	0	0	0	1		
	CDH23.3	216	81	78	11	0	0	0	0	0	0	0	6	41		
	CDH26.7	54	24	23	1	0	0	0	0	0	0	0	1	6		
(14) Wind	WSAvg	6.8	8.7	7.7	6.8	6.6	5.1	5.3	4.9	5.6	6.2	7.5	8.6	8.9		
(15) Precipitation	PrecAvg	579	26	22	39	51	86	79	82	62	40	32	30	27		
	PrecMax	951	104	76	116	184	344	224	223	150	150	147	123	78		
	PrecMin	277	0	0	2	2	9	1	11	11	2	0	1	0		
	PrecStd	149	23	17	26	40	58	51	52	31	28	26	23	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.9	30.0	24.8	20.0	14.9	10.2	10.3	11.8	17.2	19.8	23.9	27.4		
		MCWB	16.5	15.4	14.2	12.0	10.3	6.9	7.2	7.1	8.3	10.5	12.7	15.7		
	2%	DB	25.1	25.9	21.3	17.0	12.4	8.9	8.7	9.8	13.7	16.9	20.0	23.9		
		MCWB	14.5	14.5	12.4	10.9	8.9	6.3	6.4	6.4	7.3	9.2	11.0	14.0		
	5%	DB	22.1	22.2	18.9	14.9	10.9	7.8	7.1	8.5	11.8	14.7	16.9	20.7		
		MCWB	13.4	13.5	11.6	10.1	8.0	5.6	5.1	5.5	6.8	8.6	10.2	12.6		
	10%	DB	19.6	19.1	16.8	13.1	9.2	6.5	6.0	7.1	10.1	12.9	14.8	18.0		
		MCWB	12.6	12.3	10.9	9.1	6.9	4.6	4.1	4.6	5.9	7.7	9.3	11.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.7	16.5	15.3	13.2	10.9	7.7	7.8	8.4	9.5	11.4	13.2	16.0		
		MCDB	28.2	26.6	22.2	17.9	13.7	9.6	9.8	10.4	15.5	17.9	21.6	26.0		
	2%	WB	15.0	15.1	13.4	11.9	9.5	6.5	6.5	6.9	8.1	9.8	11.8	14.3		
		MCDB	23.5	23.9	19.3	15.7	11.9	8.3	8.2	9.2	12.6	15.3	18.6	22.4		
	5%	WB	13.9	13.9	12.2	10.5	8.3	5.5	5.3	5.7	7.1	8.9	10.6	12.9		
		MCDB	20.9	20.9	17.5	14.2	10.6	7.2	6.9	7.8	10.8	13.9	16.2	19.8		
	10%	WB	12.7	12.8	11.2	9.4	7.2	4.6	4.1	4.8	6.2	8.0	9.5	11.7		
		MCDB	18.6	18.4	16.2	12.7	9.2	6.2	5.6	6.8	9.6	12.4	14.2	17.3		
(35) Mean Daily Temperature Range	MDBR		11.2	11.8	11.7	10.2	7.8	6.7	6.7	7.6	9.4	10.2	10.4	10.7		
	5% DB	MCDBR	18.5	19.5	19.2	14.6	9.6	7.3	7.1	9.1	13.0	14.8	15.7	17.4		
		MCWBR	9.6	10.0	11.2	9.5	6.9	5.5	5.2	6.2	7.7	8.6	8.8	9.3		
	5% WB	MCDBR	15.2	15.9	15.0	12.7	8.4	6.4	6.4	8.0	11.3	12.1	13.3	15.1		
		MCWBR	8.3	8.9	9.1	8.8	6.3	5.1	4.9	5.7	7.0	7.3	7.8	8.5		
(40) Clear-Sky Solar Irradiance	taub		0.306	0.308	0.297	0.286	0.273	0.263	0.261	0.278	0.294	0.301	0.298	0.308		
	taud		2.571	2.569	2.593	2.616	2.603	2.564	2.548	2.575	2.547	2.539	2.566	2.545		
	Ebn at Noon		1016	984	948	882	815	780	816	877	934	981	1018	1020		
	Edh at Noon		102	96	84	68	56	51	57	69	86	98	102	106		
(44) All-Sky Solar Radiation	RadAvg		7.11	6.00	4.32	2.76	1.55	1.11	1.32	2.09	3.41	5.11	6.49	7.16		
	RadStd		0.38	0.48	0.27	0.15	0.15	0.12	0.12	0.19	0.31	0.36	0.29	0.57		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page